

Ashish Molakalapalli

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EDUCATION

Master of Science in Computer Science

Texas A&M University

Aug 2025 – May 2027

College Station, TX

- GPA: 3.833 / 4.0 | Graduating **May 2027** | Coursework: Operating Systems, Analysis of Algorithms, Machine Learning, Artificial Intelligence, NLP Foundations & Tech, Information Storage & Retrieval
- Focus: Scalable Systems, Cloud Infrastructure, API Design, Distributed Computing

Bachelor of Technology in Computer Science & Engineering

IIIT Kottayam

Dec 2021 – Apr 2025

India

- GPA: 9.34 / 10.0 | Selected Research Scholar, IoT Cloud Research Group – Top 5 of 150+ applicants

INDUSTRY EXPERIENCE

Software Engineering Intern – Security Platform

ServiceNow

May 2025 – Aug 2025

San Diego, CA

- Designed and shipped a modular 14-tool service layer (JavaScript, Now Assist Skill Kit) orchestrating parallel GlideRecord queries, REST callouts, and scheduled-job introspection across ServiceNow's production security infrastructure with zero manual intervention.
- Built a schema-driven analysis pipeline with a 14-field JSON contract and confidence-based escalation logic; integrated AWS Bedrock as the inference backend, achieving consistent structured output across all instance types in production.
- Engineered a real-time alerting service delivering adaptive-card health summaries to Microsoft Teams via webhooks, replacing a manual multi-instance review process and cutting incident response latency from hours to minutes.
- Resolved cross-environment deployment conflicts during PDI-to-SecurityDev migration by auditing update sets, isolating conflicting records, and establishing a repeatable porting protocol adopted for subsequent releases.

Software Engineering Intern

Shipsy

Sep 2024 – Jul 2025

Gurugram, India

- Engineered event-driven automation workflows integrating Slack, Gmail, and GitHub APIs to process 10K+ requests per day for DTDC; implemented request batching and async processing, reducing manual triage by 35% and cutting average API latency by 40%.
- Built a prototype knowledge graph service with structured entity linking and full-text search, improving internal documentation query response time by 40% under concurrent load.
- Designed and shipped an autonomous standup transcription pipeline (AI Scrum Agent) with retry logic, idempotency guarantees, and Jira write-back; won Best Idea Award at Shipsy Hackathon 2025.

Software Developer Intern

UST Global

Jun 2024 – Aug 2024

Kochi, India

- Architected SmartChat as a modular microservices system; exposed NLP inference pipelines via production-grade REST endpoints with structured request validation, error handling, and response observability.
- Optimized request routing and introduced query-time caching, reducing average resolution latency by 30% and improving end-to-end accuracy by 25% under sustained load.

Software Intern

Nav Tech

Dec 2023 – Feb 2024

Hyderabad, India

- Implemented low-latency ingestion APIs for UWB hardware data streams using async I/O and connection pooling, reducing average response time by 45%.
- Introduced health-check endpoints and alerting hooks integrated with the monitoring stack, sustaining 99.9% service uptime across the device fleet.

PROJECTS

Real-Time Job Ingestion and Deduplication Platform | Python, GitHub Actions, Concurrent I/O, REST APIs, JavaScript

- Built a concurrent ingestion service polling 950+ company applicant-tracking boards across 6 platform APIs (Greenhouse, Lever, Ashby, SmartRecruiters, Workable, Workday), sweeping 52K+ postings per run with a bounded thread pool and per-source fault isolation.

Scalable Task Management Platform | *React, Node.js, WebSockets, PostgreSQL, AWS EC2/RDS, Docker, JWT*

- Implemented JWT authentication, RBAC, and query indexing strategies to handle high-concurrency workloads without lock contention; load-tested to 500 concurrent users with sub-200ms P99 latency.

Financial Document Intelligence: Production RAG System | *RAG, GPT-4o, FAISS, BM25 Hybrid Search, Python, Flask, Docker, AWS EC2*

- Containerized the service with Docker and deployed on EC2; exposed low-latency REST endpoints with per-query source attribution, reducing analyst document retrieval time by 50% in production use.

E-Commerce Order Management Platform | *React, Node.js, MySQL, AWS ALB/RDS, Auto Scaling, Docker*

- Orchestrated checkout and order APIs with rate limiting, idempotency keys, and request validation; modeled the relational schema on RDS MySQL ensuring transactional integrity under concurrent load.

TECHNICAL SKILLS

Languages: Python, JavaScript (Node.js, React), C++, Java, SQL, Bash

Backend & Systems: REST APIs, Microservices, WebSockets, Async I/O, Event-Driven Architecture, Rate Limiting, Message Queuing, Concurrency

Databases: PostgreSQL, MySQL, Query Optimization, Indexing, Transactions, FAISS (vector search), Redis

Cloud & DevOps: AWS (EC2, RDS, S3, ALB, SageMaker, Bedrock, Auto Scaling), Docker, Git, CI/CD, GitHub Actions, SLURM

Tools & Platforms: ServiceNow (Skill Kit, GlideRecord, Update Sets), Flask, n8n, Weights & Biases, Jira, Pandas, NumPy

CS Fundamentals: Distributed Systems, System Design, Data Structures & Algorithms, OS Concepts, Networking, Concurrency

RESEARCH EXPERIENCE

Research Assistant, NLP Lab

Feb 2026 – Present

Texas A&M University

College Station, TX

- Built an automated ingestion and feature-engineering pipeline converting raw SLURM telemetry from 941 profiled HPC training jobs across 9 NVIDIA GPU configurations (A100, A40, RTX, T4, and multi-GPU) into a structured 37-feature dataset used for downstream model training.
- Extended the benchmark with a 1,884-run LLM inference dataset capturing prefill/decode split timing, TTFT, TPOT, and KV-cache pressure across 22 models and 4 GPU types.
- Trained gradient-boosted and neural regressors on the resulting dataset, reaching 23.7% MAPE ($R^2 = 0.881$) on training-job runtime and 8.8% MAPE ($R^2 = 0.982$) on inference latency.
- Instrumented distributed training runs across DistributedDataParallel, DataParallel, and DeepSpeed ZeRO stages 1–3 to capture multi-GPU scaling behaviour under mixed precision.
- **Accepted to Findings of EMNLP 2026** – co-authored survey on deep learning job runtime prediction for GPU cluster scheduling, built on the 941-job benchmark above, in collaboration with XPerf and Texas A&M University.

Graduate Researcher, NLP Lab

Sep 2025 – Feb 2026

Texas A&M University

College Station, TX

- Designed an 8-stage automated data processing pipeline converting P&ID engineering schematics to structured JSON graph outputs; integrated YOLOv8 for component detection, PaddleOCR with EasyOCR for multi-scale text extraction, and HDBSCAN with UMAP for symbol clustering.
- Containerized the pipeline as a REST API service with structured JSON output contracts, enabling integration with downstream LLM reasoning and knowledge graph construction systems.

ACHIEVEMENTS & PUBLICATIONS

- **Findings of EMNLP 2026 (Accepted)** – *Deep Learning Job Runtime Prediction for GPU Cluster Scheduling*; co-authored survey on deep learning job runtime prediction for GPU cluster scheduling, in collaboration with XPerf and Texas A&M University, grounded in a 941-job, 37-feature benchmark across 9 GPU configurations plus a 1,884-run inference dataset.
- **Hackathon Winner** – Best Idea Award, Shipy Hackathon 2025 (40+ teams) for an autonomous standup-to-Jira pipeline; National Finalist, Smart India Hackathon competing against 500K+ participants across institute, regional, and national rounds.
- **AI/ML Club Sub-Lead**, IIIT Kottayam (2024) – Directed 8 ML workshops and 3 hackathons; grew active membership by 40% and mentored 30+ students. **IEEE Event Organizer** (2023) – Hosted the IoT Cloud Research Symposium with 200+ attendees across 6 research sessions.
- **CBSE Merit Certificate** – Perfect score 100/100 in Mathematics (Grade 10), awarded to the Top 0.1% of all national candidates.